

Authorized to Use, Forbidden to Influence: Purpose-Selective AI for Financial Decision Systems

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Abstract

Financial organizations are often authorized to hold confidential attributes for one purpose while being prohibited from using them for another. A purpose-selective system must therefore satisfy two properties at once: preserve the authorized utility of the attribute (it may *use* it) and prevent its cross-purpose influence (it must not *act on* it). We define Purpose-Selective Bounded Execution (PSBE) and FinBoundBench, a paired benchmark that measures both properties at the decision level against a natural-decision floor. FinBoundBench constructs counterfactual variants of the same financial case in which a confidential field is authorized, prohibited, or masked, and asks a deployed model to make the same routing or review decision. In a preregistered, frozen, and independently re-computed confirmatory study on two model families and two financial tasks, prohibited-visible data changed decisions 96–100% of the time (far above the 0–4% floor), masking the field suppressed that influence to at or below the floor in every condition, and purpose-annotated governed execution retained the full authorized utility gain (utility-retention ratio of 1.0 in both studies). The results show that decision-level influence, not transmission, is the quantity that must be measured and controlled, and that a governed execution adapter can achieve purpose selectivity without degrading authorized performance.

Keywords

purpose limitation, data minimization, financial AI, LLM governance, benchmarking, decision influence

1 Introduction

Consumer-finance workflows increasingly route individual decisions through large language models (LLMs): a support message is routed to hardship review, a flagged transaction is cleared or escalated. The attributes consulted in these decisions are purpose-bound by law and by contract. Under the principle of purpose limitation and data minimization [2, 10], a lender may hold a hardship flag to evaluate repayment-assistance eligibility (authorized purpose) while being prohibited from using the same flag to decide service priority (prohibited purpose). The European AI Act similarly treats credit and insurance scoring as high-risk applications with strict purpose boundaries [11].

Modern access-control and privacy machinery addresses where data may *go* (transmission), not what it may *do* at the decision. Confidential computing [12, 19, 23], usage control [13], and purpose-based access control (PBAC) [4] govern release. Prompt-injection defenses [8, 9] protect the prompt. But a model that is allowed to *receive* a field for an authorized purpose can still *act on* it under a prohibited purpose: the field is visible in the prompt, the model weighs it, and the decision drifts. Transmission-level controls neither measure nor prevent this drift.

We make three contributions.

- (1) **Definition.** We formalize *Purpose-Selective Bounded Execution (PSBE)* as two measurable decision-level properties over a confidential attribute: the *authorized utility* it preserves for the permitted purpose, and the *prohibited influence* it exerts under other purposes, measured relative to the system’s own natural-decision floor.
- (2) **Benchmark.** We introduce *FinBoundBench*, a paired benchmark of synthetic-but-realistic consumer-finance cases in which the same confidential field (e.g., a hardship flag, a fraud flag) is embedded in purpose-labeled decision tasks. Each case is rendered into counterfactual variants (authorized, prohibited, masked), so a single model serves as its own control.
- (3) **Evidence.** We report a preregistered, frozen, and independently re-computed confirmatory study across two model families (DeepSeek V4 Pro; Kimi K3) and two financial tasks (hardship-support routing; fraud review). Prohibited-visible data changed decisions on 96–100% of cases (floor: 0–4%); masking the field suppressed influence to at or below the floor in every condition; and governed execution with a declared purpose retained the full authorized utility gain (utility-retention ratio of exactly 1.0 in both studies).

Our framing is deliberately modest. The study covers two model-task pairs over semi-synthetic signals; it does not claim formal guarantees, trusted-hardware isolation, or adversarial robustness (Section 8). Its contribution is a measurement methodology that isolates the quantity that matters—*decision influence relative to a floor*—together with confirmatory evidence that a governed execution adapter can be purpose-selective without an authorized-utility penalty.

2 Related Work

Purpose-based access control. PBAC binds access permissions to the purpose of a data request [4]; variants attach privacy-aware purpose constraints at storage and query time. Usage control extends this with obligations and conditions at data-consumption time [13]. These systems govern *release*: they decide who may read what, for which declared reason. They do not observe the downstream decision, and they cannot detect a model weighting a field it was permitted to read for one purpose while acting under another.

Contextual integrity. Nissenbaum’s contextual integrity treats appropriateness of data flow as the central privacy criterion [15], and Barth et al. derive a formal language for context-relative norms [1]. CI-Work operationalizes flow norms as runtime checks for applications [5]. These lines treat *flow* as the regulated object. Our benchmark treats *influence* as the regulated object: a flow can be perfectly legal and still causally tilt a prohibited decision.

LLM privacy benchmarks. ToolPrivacyBench audits whether agent tools leak or mishandle private data in tool calls [17]; CONFAIDE generates contextualized private-data scenarios [22]; PrivaCI-Bench builds role-negotiation tasks for privacy context inference [16]. These benchmarks ask whether private data is *exposed* to a downstream consumer or tool. FinBoundBench instead asks whether a decision *changes* when the data’s purpose flips—the transmission may be identical.

Fairness and bias influence. Counterfactual fairness asks whether predictions are invariant under counterfactual changes of protected attributes [14]. Fin-Bias measures which prompt features drive financial-advice behavior under instruction changes [20]. Both are methodologically close to us: they vary an input while holding the case fixed. But they target *bias* (protected groups, advice consistency), not *purpose authorization*; neither reports against a decision floor, and neither is preregistered with frozen artifacts. IFC-ML analyzes information-flow control for ML pipelines [21] at the system level rather than the decision level.

Regulatory framing. GDPR Articles 5(1)(b) and 5(1)(c) ground purpose limitation and minimization [10]; the EU AI Act Article 6 Annex III classifies creditworthiness and insurance-risk systems as high-risk, tying purpose discipline to deployment obligations [11]. Work on reviving purpose limitation in data-driven systems [2] argues that technical enforcement lags legal obligation—the gap this paper addresses with a measurable property.

Position. The closest conceptual relative is ToolPrivacyBench (transmission of private data) and the closest methodological relative is Fin-Bias (input perturbation). To our knowledge no existing benchmark measures, on the same confidential signal, the *joint* requirement of authorized utility retention and floor-relative prohibited influence, under a preregistered protocol with frozen, independently re-computed evidence.

3 Definitions and Threat Model

3.1 Purpose and authorization

Let a *case* be a tuple of observable attributes x (public features) and confidential attributes z . A *purpose* p is a task label under which a system may act on a case (e.g., `HARDSHIP-SUPPORT-ROUTING`, `FRAUD-REVIEW`). For a confidential attribute z_i , an authorization mapping $\text{perm}(z_i, p) \in \{\text{AUTH}, \text{PROH}\}$ states whether the organization may use z_i when acting under purpose p . A *decision function* $d(\tilde{x}, p) \rightarrow \mathcal{A}$ maps the rendered input $\tilde{x}$ (a prompt over visible attributes) and the purpose to an action in $\mathcal{A}$.

3.2 The PSBE property

For a fixed case and fixed public features, consider three renderings of the same confidential attribute z_i : visible ($\tilde{x}^+$), masked ($\tilde{x}^-$), and the identity of the model’s own nondeterminism (two identical renderings). We define

- **Authorized utility** $U(p)$: decision quality on the authorized task, e.g., balanced accuracy (BACC) against a ground-truth label.
- **Influence** $I(p)$: the rate at which a decision *changes* when a masked attribute becomes visible, under purpose p : $I(p) = \Pr[d(\tilde{x}^+, p) \neq d(\tilde{x}^-, p)]$.

- **Natural-decision floor** F : the rate at which the decision changes when the *identical* input is rendered twice, $F = \Pr[d(\tilde{x}, p) \neq d'(\tilde{x}, p)]$ over repeated calls. Nondeterministic LLM sampling makes $F > 0$.

Definition (PSBE). A system is *purpose-selective* on attribute z_i with margin δ and retention gate ρ if, for every authorized purpose p^+ and prohibited purpose p^- :

$$\begin{aligned} I_{\text{governed}}(p^-) &\leq F + \delta && \text{forbidden to influence} \\ U_{\text{governed}}(p^+) &\geq \rho \cdot U_{\text{ungoverned}}(p^+) && \text{authorized to use.} \end{aligned} \quad (1)$$

The first conjunct bounds prohibited influence at the floor (the system is *forbidden to influence*); the second requires the authorized utility to survive governance (the system remains *authorized to use*). Both are behavioral, decision-level properties: they hold or fail at the model output, regardless of where data was stored or transmitted.

3.3 Threat model

We target *operational purpose confusion*—an organization that holds a field for a lawful purpose and feeds it to a model that acts under another purpose—not adversarial leakage or prompt injection. Concretely: the threat model includes negligent or erroneous deployment (field included when it should not be; no runtime guard), and honest-but-curious model providers (the operator controls the model call through its own governed adapter). It explicitly excludes: (i) prompt-injection attacks that attempt to exfiltrate the field; (ii) a hostile provider that colludes to train on the field (no training-data protection is claimed); (iii) formal non-interference guarantees (we measure, we do not prove). We additionally require that the deployed system never automatically falls back to an ungoverned path when the governed path fails, because a fallback would silently bypass the property in Equation (1).

4 FinBoundBench

4.1 Data

FinBoundBench uses consumer-finance records from public sources: mortgage application records from the 2024 Home Mortgage Disclosure Act (HMDA) data for the District of Columbia [7] and complaint records from the Consumer Financial Protection Bureau (January 2024) [6]. No personal data is used. From each record we synthesize one confidential attribute z_i (a hardship flag for the routing task; a high-risk fraud flag for the review task; all such fields are prefixed `SYNTHETIC-` and are not present in the source records) and a ground-truth label y :

$$y = g(x) + \beta z_i + \varepsilon, \quad (2)$$

where g is a public-feature scoring function, β controls the authorized-signal strength of z_i , and ε is irreducible noise. A separate *prohibited rule*—unknown to the model, e.g., a policy that prohibits using the hardship flag for service priority—defines which variant of the label is correct under each purpose: the label is *invariant* across purpose variants, so any decision change caused by the field’s purpose assignment is attributable to influence, not to a changed ground truth.

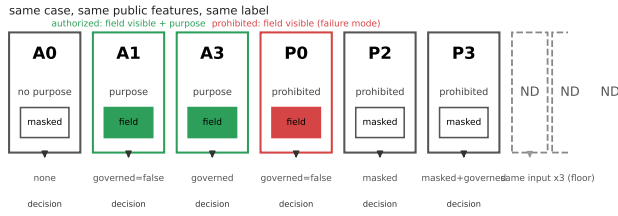


Figure 1: The seven purpose-paired renderings of a single case in FinBoundBench. The confidential field is green when authorized, red when visible under a prohibited purpose (the failure mode), and gray when masked. The natural-decision floor (ND) renders the identical input three times.

Signal strength is calibrated so that a public-features-only model achieves BACC in $[0.55, 0.85]$ and the authorized gain of the confidential field is at least 0.08, with class prevalence in $[0.30, 0.70]$ (sensitivity gates). A held-out *confirmatory split* is sampled with fixed seed after calibration and locked by content hash. The dev split used for harness development and for the figures in this paper contains $\sim 1,000$ pairs per signal; the confirmatory split used for all reported tests contains 100 pairs (primary study) and 120 pairs (replication study).

4.2 Purpose-paired conditions

Each case is rendered into seven conditions, keeping public features and labels identical:

Condition	Rendering
A0	no purpose, no metadata, no context (baseline)
A1	full purpose and context, field visible, ungoverned
A3	full purpose, field visible, governed execution (PSBE)
P0	prohibited purpose, field visible (the failure mode)
P2	prohibited purpose, field masked deterministically
P3	prohibited purpose, field masked by the governed contract
ND	identical input rendered $\times 3$ (floor estimate)

The key contrast pairs are authorized (A0 vs. A1, A1 vs. A3) and prohibited (P0 vs. P2, P0 vs. P3, P3 vs. ND). Because the same case is rendered across all conditions, each model serves as its own control: differences are within-case, and sample size is in *pairs*, not independent cases. Figure 1 shows the seven renderings of one case.

4.3 Metrics

- **Balanced accuracy (BACC):** per-condition decision accuracy, macro-averaged over classes. Authorized utility gain $G = \text{BACC}_{A1} - \text{BACC}_{A0}$.
- **Utility-retention ratio (AUR):** the fraction of the authorized gain retained by governed execution, $\text{AUR} =$

G_{gov}/G , where $G_{\text{gov}} = \text{BACC}_{A3} - \text{BACC}_{A0}$. $\text{AUR} = 1$ means governed execution retains the entire purpose benefit.

- **Unjustified influence rate (UIR):** the fraction of pairs on which the decision changes between a masked and a visible rendering (P0 vs. P2, P0 vs. P3). The *floor* F is the change rate between identical ND renderings. *Net unjustified influence* is $\text{NetUI} = \text{UIR}_{P0} - F$.
- **Availability:** the fraction of governed calls that complete without provider or routing failure; **evidence integrity:** every governed call records an append-only, hash-bound event (Section 5).

4.4 Leaderboard objective

FinBoundBench is also the evaluation harness for a public competition (ICAF 2026 Challenge track), in which the ranking is lexicographic: constraint first ($\text{NetUI} \leq 0.05$, zero unauthorized-action violations, availability ≥ 0.95), then authorized utility ($\text{AUR} \geq 0.80$), then efficiency. The confirmatory results in this paper use the same metrics and conditions; a degenerate-strategy audit confirms the ranking is not gameable—e.g., an ALWAYS-USE-FULL-DATA submission achieves perfect accuracy but violates the constraint maximally, while random output sits at the floor. The paper’s contribution does not depend on the competition; the protocol is the shared artifact.

5 PSBE-Runtime

PSBE-Runtime is a governed execution adapter that sits between an application and a model API, implementing the GOVERNED conditions (A3, P3) of Equation (1). It is deliberately thin: it does not fine-tune, post-process, or re-rank model output, and it never rewrites decisions. It enforces *purpose* at the boundary.

- (1) **Purpose contract.** Each call declares a purpose p from a fixed registry. A machine-readable contract maps p to the set of fields it may see (AUTHORIZED) and the set it may not (PROHIBITED).
- (2) **Projection.** The runtime renders the prompt from a projection of the case: prohibited fields are excluded before any token reaches the model. For the authorized rendering (A3) the confidential field is transmitted and annotated with its declared purpose; for the prohibited rendering (P3) the field is excluded by the same contract.
- (3) **Purpose-annotated context.** The authorized rendering attaches the declared purpose and the decision constraints (which actions are permitted), so the model conditions on the purpose that authorizes the field—the mechanism by which the benchmark’s A1 condition turns into governed behavior.
- (4) **Output validation.** Actions are validated against the permitted action set for p ; disallowed actions are rejected and recorded, never silently remapped.
- (5) **Evidence chain.** Every call appends an event (purpose, contract version, input hash, output hash, timestamp) to an append-only, hash-linked log, enabling tamper-evident reconstruction of what any decision saw.

- (6) **No fallback.** If the governed path fails (transport, timeout), the call is recorded as a failure and excluded under a pre-registered failure taxonomy; the runtime never falls back to an ungoverned path, because fallback would reintroduce the influence the contract exists to prevent.

The runtime is deployed as a library adapter around the model API; the same adapter serves both the confirmatory study and the benchmark's GOVERNED conditions, so the measured property is exactly the production property. Its cost is a single projection and a log append per call; measured overhead is negligible relative to model latency and is not a factor in the results below.

6 Experimental Setup

6.1 Pre-registration

The protocol, hypotheses, gatekeeping rules, power analysis, metric definitions, seed, and failure taxonomy were written and frozen before any confirmatory execution, and the confirmatory split was locked by content hash at the same freeze. All artifacts are content-addressed (SHA-256); results were frozen at fixed timestamps and never regenerated after the freeze.

Hypotheses (one-sided unless noted):

ID	Claim	Status
H1	Authorized gain $G = \text{BACC}_{A1} - \text{BACC}_{A0} > 0$	confirmatory
H2	$\text{AUR} \geq 0.80$	confirmatory
H3	$\text{NetUI} = \text{UIR}_{P0} - F > 0$	confirmatory
H5	UIR suppression: $\text{UIR}_{P2}, \text{UIR}_{P3} < \text{UIR}_{P0}$	confirmatory
H6	P3 within $\delta = 0.05$ of floor	confirmatory
H7	P2 = P3 (equivalence, TOST, $\delta = 0.05$)	report-only
H4	$P1 < P0$	not testable (P1 outside registered minimal set)

Gatekeeping was fixed-sequence: chain 1 (H1 $\rightarrow$ H2) gates authorized-utility claims; chain 2 (H3 $\rightarrow$ H5 $\rightarrow$ H6) gates influence claims; H7 is reported without gating. Power analysis fixed $n = 100$ pairs for the primary study (power ≥ 0.95 at a conservative gain of 0.167) and $n = 120$ for the replication study. Significance: paired cluster bootstrap (5,000 iterations) with one-sided p ; McNemar exact test on discordant pairs; TOST with $\delta = 0.05$ for equivalence. The execution budget was fixed at EUR 40; routing policy fixed candidate models *before* execution.

6.2 Admission

Candidate (model $\times$ task) lanes passed a pre-registered eligibility gate on a calibration split: sensitivity of the public-only baseline ($\text{BACC} \in [0.55, 0.85]$), authorized gain ≥ 0.08 , and class prevalence in $[0.30, 0.70]$. This gate is why the results are reported per (model, task) pair and not pooled. An earlier protocol generation (v3) was withdrawn after its evaluation used test-double substitutes (silent-compromise rate 0.2426, availability 0.9857, AUR 0.9341) that were found invalid for live claims; v4 mandates live-event metrics and

Table 1: Primary study (DeepSeek V4 Pro, hardship-support routing).

Metric	Value
BACC A0 (no purpose)	0.32
BACC A1 (full purpose)	0.71
BACC A3 (governed)	0.71
UIR P0 (prohibited, visible)	1.00
UIR P2 (masked, deterministic)	0.00
UIR P3 (masked, contract)	0.00
ND floor	0.00

Hyp.	Decision	Point	p (1-sided)
H1 gain	PASS	0.39	0.00020
H2 AUR	PASS	1.0	—
H3 NetUI	PASS	1.00	0.00020
H5 suppression	PASS	−1.00	0.0
H6 P3 vs. floor	PASS	−0.00	0.0
H7 P2 = P3	EQUIV.	0.00	$p_{\text{TOST}} 0.0$

the gates above. We report this history because it motivated the protocol.

6.3 Execution

Two confirmatory studies ran on frozen splits with the fixed seed: the primary study (DeepSeek V4 Pro on hardship-support routing; 100 pairs; 1,300 calls; 100% provider success; EUR 2.66) and the replication study (Kimi K3 on fraud review; 120 pairs; 1,560 calls; 99.0% success). The replication study required seven execution windows: windows 1–7 failed provider-side (transport errors and timeouts under a pre-registered failure taxonomy) and window 8 was accepted with registered failure handling; 16 transient failures were excluded per the taxonomy, leaving 114 pairs complete on all six conditions. The replication cost was not recorded in the frozen artifacts. Both studies ran the same conditions (A0, A1, A3, P0, P2, P3, and ND $\times$ 3), the same labels, and the same governed adapter (PSBE-Runtime) as the A3/P3 conditions.

7 Results

7.1 Primary study (DeepSeek V4 Pro / hardship-support routing)

100 pairs, 1,300 calls, 100% provider success, EUR 2.66.

Table 1 summarizes the primary study. Declaring purpose raised balanced accuracy from 0.32 (no purpose) to 0.71 (H1 gain 0.39, 95% CI [0.21, 0.56], $p = 0.00020$); governed execution retained the entire gain (AUR = 1.0, 95% CI [1.0, 1.0], H2). Prohibited-visible data changed the decision on *every* pair ($\text{UIR}_{P0} = 1.00$ versus a floor of 0.00; H3 net influence 1.00, 95% CI [1.00, 1.00], McNemar exact $p = 1.58 \times 10^{-30}$), and both masking variants suppressed that influence exactly (UIR = 0.00; H5, H6).

7.2 Replication study (Kimi K3 / fraud review)

120 pairs, 1,560 calls, 99.0% provider success.

Table 2: Replication study (Kimi K3, fraud review).

Metric	Value
BACC A0 (no purpose)	0.625
BACC A1 (full purpose)	0.7333
BACC A3 (governed)	0.7333
UIR P0 (prohibited, visible)	0.9646
UIR P2 (masked, deterministic)	0.00
UIR P3 (masked, contract)	0.0083
ND floor	0.0375

Hyp.	Decision	Point	p (1-sided)
H1 gain	PASS	0.1083	0.00200
H2 AUR	PASS	1.0	—
H3 NetUI	PASS	0.9292	0.00020
H5 suppression	PASS	-0.9646 / -0.9558	0.0
H6 P3 vs. floor	PASS	-0.0292	0.0236
H7 P2 = P3	EQUIV.	-0.0083	$p_{\text{TOST}} 9.98 \times 10^{-7}$

Confirmatory protocol-v4 — both studies (frozen) - H2 badge = AUR decision; H3/H5/H6 badges = chain-2 decisions

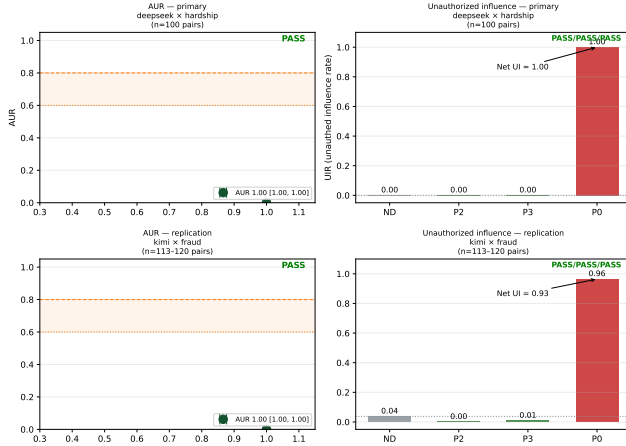


Figure 2: Headline results. Rows: primary study (top), replication study (bottom). Left panels: authorized utility (BACC) per condition, with the governed condition (A3) equal to the full-purpose condition (A1). Right panels: unjustified influence rate (UIR) per condition against the natural-decision floor; prohibited-visible data (P0) sits at or near 1.0 while masked variants (P2, P3) sit at or below the floor.

The replication (Table 2) reproduces every gate on a second model family and a second task. Purpose gain 0.1083 (95% CI [0.033, 0.175], $p = 0.00200$); AUR = 1.0; prohibited-visible influence 0.9646 against a floor of 0.0375 (net 0.9292, 95% CI [0.8805, 0.9646], McNemar $p = 3.94 \times 10^{-31}$); masking suppressed influence to 0.00 (P2) and 0.0083 (P3), with P3 statistically equivalent to the floor within the 0.05 margin (H6, point -0.0292, 95% CI [-0.0625, 0.0], $p = 0.0236$) and P2 equivalent to P3 (H7, $p_{\text{TOST}} = 9.98 \times 10^{-7}$).

7.3 Combined interpretation

Both studies passed every pre-registered gate, satisfying interpretation rule 1: *the effect reproduces across two model families and two financial tasks*. The frozen evidence was re-computed independently (fresh implementation over raw events, no shared code with the reporting scripts): 17/17 verification checks pass, including exact reproduction of every headline metric in Tables 1 and 2 and both McNemar p -values.

7.4 What the results do and do not show

The governed condition (A3) and the masked conditions (P2, P3) were equivalent in influence in both studies, and governed execution retained the full authorized gain. We make no claim that governed execution *outperforms* masking: on a single narrow field, deterministic masking (P2) and contract masking (P3) are behaviorally indistinguishable, as the H7 equivalence shows. The contribution is that the *same* adapter that improves authorized accuracy by carrying purpose context (A1 $\rightarrow$ A3) also enforces the prohibited boundary (P3) at the floor, jointly satisfying Equation (1) in every condition tested.

8 Limitations and Ethics

Scope of evidence. Two (model, task) pairs were tested. The replication study’s provider required seven execution windows before any window completed (registered failure handling); this is a fact about commercial API reliability and a warning that live-model confirmatory work is operationally fragile. The replication cost was not recorded in the frozen artifacts, a process gap we note rather than conceal.

Hypothesis H4 (a purpose-agnostic masked variant) is not testable in this design: the variant was not part of the registered minimal condition set, so no events exist. Claims are limited to the conditions actually executed.

No formal guarantees. PSBE is a measured behavioral property, not a proof of non-interference. A governed adapter can still be bypassed by prompt-injection exfiltration, by a colluding provider, or by an application that reconstructs the field from other inputs; none of these are in scope. Trusted-hardware or attestation mechanisms [3, 18] can complement, not replace, decision-level measurement.

Model drift. Model behavior changes over time and per route; a purpose-selective property confirmed today is a property of a specific model revision. The protocol’s fixed route and freeze discipline bound, but do not eliminate, this threat.

Masking equivalence. P2 and P3 were equivalent on a single narrow field; on broader confidential surfaces or with implicit correlations, a purpose-contract may leave residual leakage through correlated public features, which our design does not measure. The ND floor partially absorbs this (influence is measured *relative* to the floor), but a more powerful adversarial reconstruction remains outside the design.

Ethics and data. All records are from public regulatory disclosures (HMDA 2024, CFPB 2024) and were synthesized into confidential fields prefixed SYNTHETIC-; no personal data, no real hardship or fraud flags, and no model training occurred on any record. The benchmark’s intended use is compliance

and governance testing, not screening or scoring of individuals. We release the benchmark, protocol, and frozen evidence for independent verification; we release no prompts containing real records.

Claim discipline. Throughout this paper, influence results are reported against the measured natural-decision floor, never in absolute terms; this is the methodological correction that earlier protocol generations lacked and the property we believe generalizes.

9 Conclusion

We defined Purpose-Selective Bounded Execution as a joint, decision-level property—an attribute may be *used* under its authorized purpose and must not *influence* decisions under prohibited purposes—and built FinBoundBench to measure both halves on paired counterfactuals against the model’s own nondeterminism floor. In a preregistered, frozen, and independently re-computed confirmatory study across two model families and two financial tasks, prohibited-visible data altered 96–100% of decisions (floor 0–4%), masking the field by contract suppressed that influence to at or below the floor in every condition, and governed execution retained the entire authorized utility gain (AUR = 1.0 in both studies).

The practical reading is direct: transmission controls are not enough; a financial institution that holds a field lawfully must still measure and bound its influence on decisions it is not permitted to shape, and a governed execution adapter—purpose contract, projection, and evidence chain—achieves that bound without an authorized-utility penalty. The methodological reading is that decision-influence evaluation, with a natural-decision floor and frozen confirmatory evidence, transfers to any regulated domain in which the same data serves multiple purposes. We release the benchmark, protocol, and frozen evidence to support independent replication, and we hope FinBoundBench becomes the first of many purpose-selective benchmarks.

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